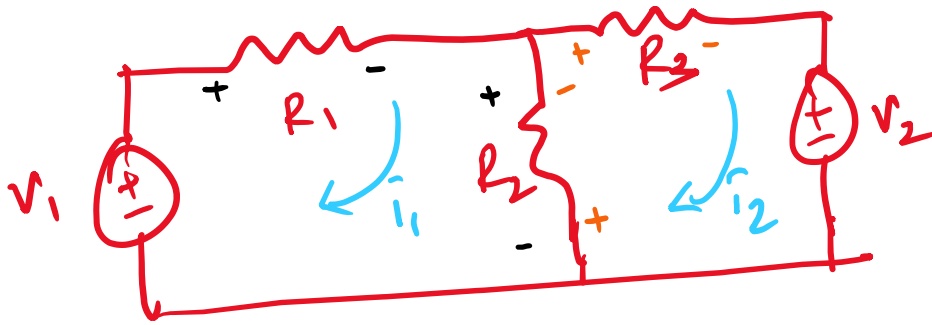


Mesh Analysis

KVL to find mesh currents.

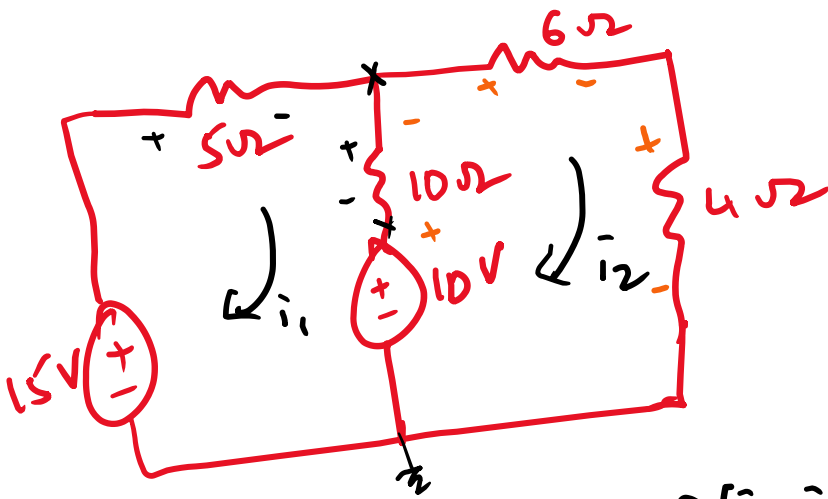
1. Identify the number of meshes.
2. Assign mesh currents in each mesh (clockwise or counter clockwise)
3. Apply KVL to each of the 'n' meshes.
4. Solve 'n' simultaneous eqns to obtain mesh currents.



$$\text{KVL @ mesh 1 : } V_1 - i_1 R_1 - (i_1 - i_2) R_2 = 0$$

$$\text{KVL @ mesh 2 : } -(i_2 - i_1) R_2 - i_2 R_3 - V_2 = 0$$

$$\text{Assume } i_2 \text{ is c-c : } V_2 - i_2 R_3 - (i_2 + i_1) R_2 = 0$$

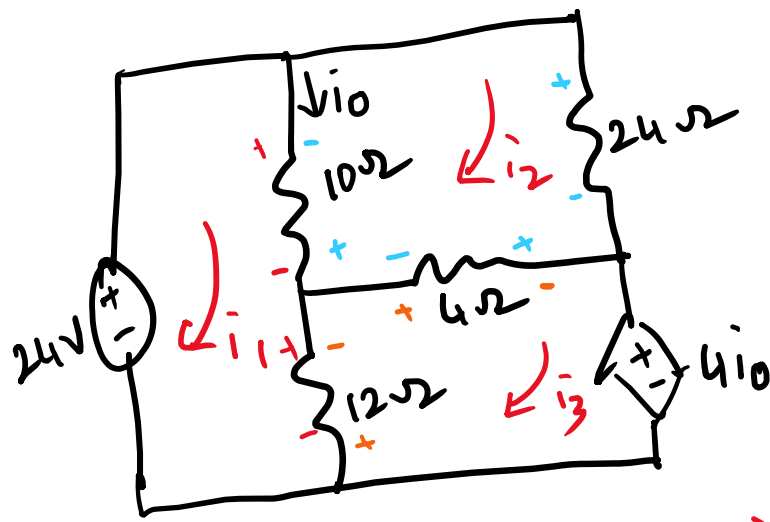


Find the currents through all the resistors using mesh analysis.

$$\text{Mesh 1 : } 15 - 5i_1 - 10(i_1 - i_2) - 10 = 0$$

$$\text{Mesh 2 : } 10 - 10(i_2 - i_1) - 6i_2 - 4i_2 = 0$$

$$i_1 = i_2 = 1\text{A}$$



Find i_0 using mesh analysis.

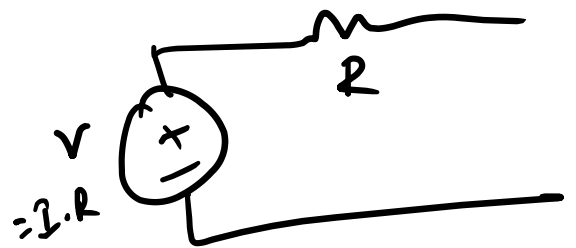
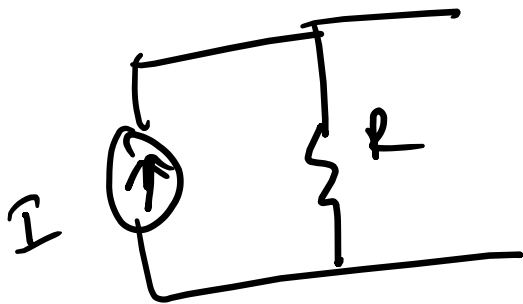
$$i_0 = i_1 - i_2$$

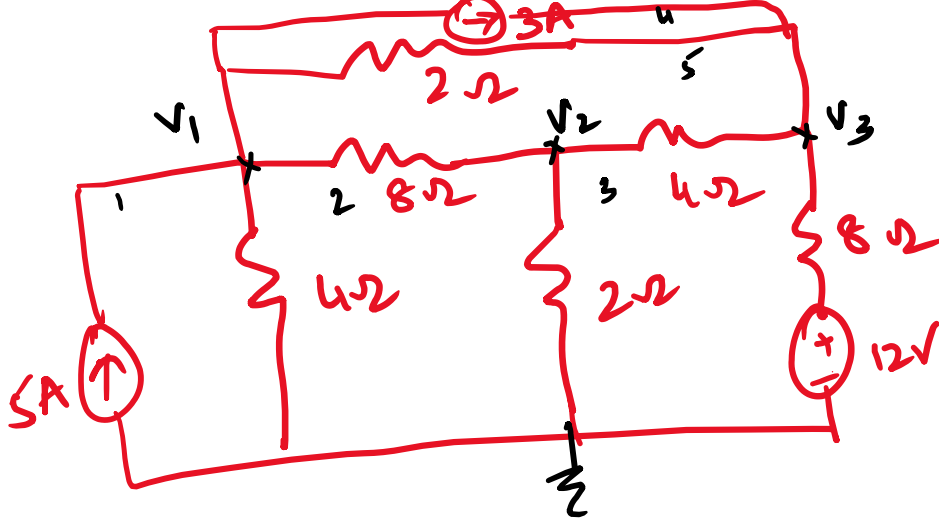
Mesh 1: $24 - 10(i_1 - i_2) - 12(i_1 - i_3) = 0$

Mesh 2: $-24i_2 - 4(i_2 - i_3) - 10(i_2 - i_1) = 0$

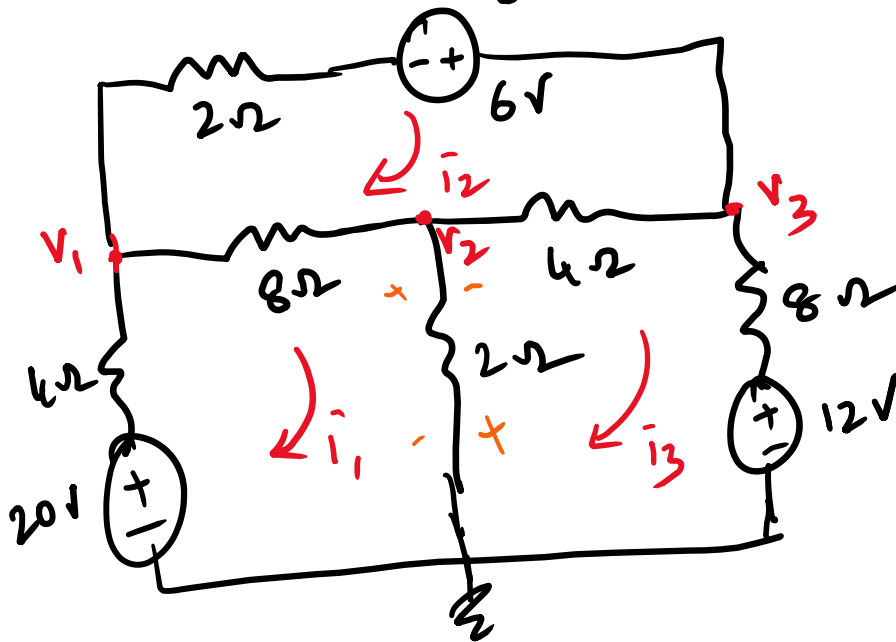
Mesh 3: $-4(i_1 - i_2) - 12(i_3 - i_1) - 4(i_3 - i_2) = 0$

Source transformation.





Find V_1, V_2, V_3
Using mesh analysis.



$$\textcircled{\uparrow} \rightarrow \textcircled{\begin{smallmatrix} + \\ - \end{smallmatrix}}$$

$$\textcircled{\rightarrow} \rightarrow \textcircled{\begin{smallmatrix} + \\ - \end{smallmatrix}}$$

Mesh 1: $20 - 4i_1 - 8(i_1 - i_2) - 2(i_1 - i_3) = 0$

2: $6 - 4(i_2 - i_3) - 8(i_2 - i_1) - 2i_2 = 0$

3: $-12 - 2(i_3 - i_1) - 4(i_3 - i_2) - 8i_3 = 0$